

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site T -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, free of modelling and shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's and the lattice kernel at normal incidence is in closed form; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying the first moment of its internal field converges at fourth order, with leading error $-(kd)^4/720$, for incident P, SV and SH

waves at normal and oblique incidence, and its spectral lattice sum needs no Ewald splitting. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. Every result is derived symbolically and verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the

17 single voxel's response is matched to cancel it, refinement converges to the
18 wrong answer without revealing it.

19 *Why a homogeneous layer.* This paper therefore starts from the simplest
20 configuration in which a voxel scheme runs in full (every voxel, every coupling,
21 the whole multiple-scattering series, at normal and at oblique incidence) and
22 the exact answer is nevertheless known: a layer filled with identical voxels.
23 Because cubes tile space, such a layer does not approximate a homogeneous
24 layer; it *is* one. Any difference between its reflection and transmission and
25 those of the continuous layer is therefore error of the discretisation alone, with
26 no modelling error and no shape error behind which to hide. A sphere, the
27 usual test object for volume-integral and discrete-dipole methods [1], cannot
28 give this: its error mixes the discretisation with the staircase approximation
29 of a curved surface by cubes. The layer plays the part of the patch test in
30 finite elements: reproducing a uniform medium correctly is necessary, though
31 not sufficient, for a scheme to be trusted in a heterogeneous one. What the
32 layer establishes about the scheme's two ingredients, the coupling between
33 voxels and the response of a single voxel, are properties of the ingredients,
34 which are the same in any medium: that the averaged coupling is exact, that
35 the self term cancels, and the form of the single-site error. §9 confirms that
36 they carry over to a stratified medium; a laterally heterogeneous one remains
37 to be tested.

38 *What the reader gains.* Four things that a heterogeneous test cannot provide.
39 First, a proof, exact at normal incidence, that the voxel scheme converges
40 to the continuum, and the specific ingredient, a coupling averaged over the
41 source voxel, without which it converges to the wrong answer. Second, the

42 leading error in closed form for every discretisation, so that a grid can be
 43 chosen for a required accuracy before the calculation is run. Third, a single-
 44 site T -matrix derived from first principles, with its error stated and checked
 45 against the exact solution for a sphere. Fourth, a voxel that converges at
 46 fourth order, also in a randomly stratified model (§9): for a reflection ac-
 47 curacy of 10^{-6} it allows voxels 58 times larger in each direction than the
 48 standard one (§10).

49 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 50 *space*, so a uniform region discretised into cubes is still that uniform region,
 51 whereas spheres or point scatterers can only approximate it and carry a pack-
 52 ing error of their own. Their *regular lattice* makes the interaction between
 53 cells translation invariant, so the coupling is a convolution (Toeplitz between
 54 planes, Bloch-periodic within them), and the whole system is solved by fast
 55 transforms. And a cube small compared with the wavelength has a *single-site*
 56 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
 57 approximation a voxel introduces is the internal field it assumes.

58 *Multiple scattering on a lattice.* Replacing a heterogeneous body by a lattice
 59 of small scatterers that interact through the background Green’s function
 60 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-
 61 magnetics its discretised form is the discrete dipole approximation [DDA;
 62 4, 5], whose accuracy has been studied in detail: each dipole’s polarisability
 63 must account for the local field of the surrounding lattice, which the lattice
 64 dispersion relation fixes so that an infinite lattice propagates waves as the
 65 continuum does to a given order in kd [6]; the error of the method converges
 66 at second order in the dipole spacing when shape errors are absent [1]; and

67 integrating the Green's tensor over each cell rather than sampling it at the
 68 centre improves accuracy, most of all for strong contrasts [7]. The elastic
 69 problem adds a structure the scalar and electromagnetic ones lack. A cubic
 70 voxel responds to the incident field with a force (through its density contrast)
 71 and a moment (through its stiffness contrast), so the state at a voxel has nine
 72 components (displacement and strain), and the single site's self-interaction is
 73 the elastic depolarisation of a cube, the discrete counterpart of the inclusion
 74 tensor of Eshelby [8].

75 *This paper.* The approach originates in the two-and-a-half-dimensional for-
 76 mulation of Nestor [9]: a medium divided into cells that interact through the
 77 background Green's tensor, each carrying a single-site contrast renormalised
 78 by its own near field, with the multiple-scattering series summed. This pa-
 79 per extends it to cubic voxels. §2 sets out the discrete medium and §3 the
 80 continuous layer it must reproduce. §4 shows why the point coupling fails,
 81 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
 82 single-site T -matrix from moment integrals, tests it against the exact sphere,
 83 and shows that with it the discrete system is a collocation of the continuum
 84 equation. §7 gives that collocation's second-order error in closed form, and
 85 §8 the fourth-order voxel, for incident P and S at normal and oblique inci-
 86 dence. §9 repeats the tests on a randomly stratified layer, and §10 discusses
 87 what the results imply for voxel schemes in general. Every result is derived
 88 symbolically in Mathematica and checked against an independent Python
 89 implementation or an exact solution.

2. Setting

The discrete medium. Space is tiled by cubes of side d (half-side $h = d/2$). Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates are ordered (z, x, y) with z down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

where $\Delta\mathcal{C}$ is the 6×6 contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

The laterally uniform layer. A layer of thickness D is filled with n planes of voxels, $d = D/n$, and illuminated at normal incidence. The field is then

laterally uniform, only the zero-lateral-wavenumber (specular) component of the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

the sum running over the square lattice $\mathbf{R}$ of pitch d , m the plane separation in pitches. The question is whether, and how fast, this chain converges to the continuous layer as $n \rightarrow \infty$ at fixed D .

Implementations. Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python implementation of the voxel scheme (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.

3. The continuous layer and its thin-layer expansion

At normal incidence a plane force drives a one-dimensional problem per wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission follow from continuity of displacement and traction at its faces; they agree with the stratified propagator used as the reference in this work to 2.5×10^{-8} , P and S (`ContinuumLimit_Reference.wl`).

For a layer of thickness D with modulus M_1 and density ρ_1 in a background M_0, ρ_0 , the reflection coefficient expands as

$$R = \frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] + O(D^2), \quad (5)$$

130 exact in the contrast (`ContinuumLimit_ThinLayer.wl`): to first order a thin
 131 layer is an interface across which the traction jumps by the excess inertia and
 132 the displacement by the excess compliance. At oblique incidence the same
 133 statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$ for the P-SV state
 134 $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$,
 135 which reproduces the layer's $O(D)$ reflection coefficients R_{PP}, R_{PS} to 10^{-12} ;
 136 the series truncated at $O(D^k)$ converges at exactly order k ; and a soft thin
 137 layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η tends to the linear-
 138 slip interface as $D \rightarrow 0$. These expansions are the target a single plane of
 139 voxels must reproduce order by order.

140 4. The specular sums of the point Green's tensor

141 By Poisson summation,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (6)$$

142 with $\hat{\mathbf{G}}$ the two-dimensional transform of the Green's tensor between planes.
 143 The $\mathbf{g} = 0$ term is the continuum's plane-wave term; the others are evanes-
 144 cent, decaying as $e^{-2\pi m \sqrt{p^2 + q^2}}$ independently of d . In the strain-from-moment
 145 block they carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel vol-
 146 ume does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 147 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 148 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 149 planes in slab order, the static lattice sum of the point kernel is exactly
 150 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 151 on the normal shears, per unit volume) plus a cubic-symmetric remainder

152 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel there-
 153 fore reproduces the continuum only if that remainder is cancelled by the
 154 voxel's own self term.

155 5. The tiling identity

156 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 157 $\mathbf{y}) \, d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 158 Each Poisson order in Eq. (6) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (7)$$

159 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 160 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 161 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ . Two exact statements follow.

162 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (8)$$

163 the continuum's plane-wave term times the vertical form factor. The pack-
 164 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 165 `AveragedSums.wl`).

166 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 167 is the field at the mid-plane of a uniform plate of thickness d :

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) \, d\mathbf{y}, \quad (9)$$

168 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') \, dz'$ is elementary: in the one-dimensional
 169 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$

170 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 171 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 172 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 173 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 174 with every cell integral evaluated analytically (below).

175 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 176 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (10)$$

177 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 178 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 179 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 180 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 181 and the quadrature order are raised; Eq. (10) is now the package’s kernel at
 182 normal incidence.

183 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 184 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 185 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 186 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 187 which never contain the origin, and there the two-dimensional antiderivative
 188 in the face coordinates is continuous, so the face integral is a sum over the
 189 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 190 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 191 *not* usable for repeated derivatives: its arctangent terms jump across the
 192 coordinate planes through the origin, and differentiating twice there discards

193 exactly that term.) The self term so obtained reproduces the classical closed
 194 form of the cube’s depolarisation to machine precision.

195 **6. The single site: the cube T -matrix from moment integrals**

196 Every error of a voxel scheme is made at the single site (§10), so the single
 197 site is derived here rather than imported. The classical route imports it. For
 198 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
 199 uniform load uniform, with a depolarisation tensor in closed form [8, 10, 11];
 200 its dynamic generalisation is known for spheres and ellipsoids [12, 13]. For
 201 a cube or a general polyhedron the static field of a uniform eigenstrain is
 202 also in closed form [14, 15, 16, 17], but the strain it produces is not uniform,
 203 so there is no exact uniform-strain inclusion to import. In scattering theory
 204 the same object appears as the self-consistent or effective-field closure of
 205 the volume integral equation [18, 19, 20]: a first-order (Born) scatterer [21]
 206 renormalised by the geometric series of its own near field, the step that turns
 207 a contrast into an effective contrast. Nestor [9] introduced the single site
 208 in this form, summing the series of the cell’s self integral, with the singular
 209 near field evaluated over an infinitesimal cell. The moments below evaluate
 210 the same self integral over the finite cube, as a distribution. The elastic T -
 211 matrix of Waterman [22] represents a scatterer in spherical multipoles, the
 212 natural basis for a sphere; a voxel in a lattice needs one compatible with the
 213 lattice, and the moments over the cube provide it. The construction below
 214 makes the approximation of a voxel explicit rather than implicit: it derives
 215 the cube’s T -matrix from the moments of the elastodynamic Green’s tensor
 216 over the cube, and identifies the uniform-strain closure as one truncation of

217 an exact hierarchy.

218 6.1. The closed hierarchy

219 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 220 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 221 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 222 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed
 223 system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij})u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{K}_{ij}^{rs})\partial_r \partial_s u_j = u_i^0, \quad (11)$$

$$(\delta_{pr}\delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj})\partial_r u_j = \partial_p u_i^0, \quad (12)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr}\delta_{ps}\delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs})\partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (13)$$

224 all fields evaluated at the centre. Its coefficients are six moments of the
 225 background Green’s tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (14)$$

226 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 227 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
 228 it, so it is fixed by convention. The moments are graded by (D, W) , the
 229 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
 230 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

231 *6.2. The moments as distributions*

232 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (15)$$

233 [23, 24]. Expanding g_c in powers of r reduces every moment in Eq. (14) to a
 234 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (16)$$

235 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 236 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 237 gral depends on the shape of the region excluded around the origin. This is
 238 the source of the δ -function term of the inclusion problem, which a numerical
 239 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 240 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 241 are distributional from the start, and each is moved onto the voxel's faces by
 242 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (17)$$

243 The surface term is evaluated on the faces, a distance h from the origin, where
 244 every derivative of r^m is smooth. The recursion lowers D and W together
 245 and ends on convergent integrals. The face integrals are done in closed form
 246 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (18)$$

247 which the face route satisfies exactly and a spherical excision returns as
 248 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell

249 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 250 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 251 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 252 Because every power of r is paired this way, the dynamic moments keep the
 253 even-power terms of the expansion (k^2r , r^3 , ...) as well as the odd ones.
 254 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 255 those terms, at a real $O((kh)^2)$ cost (§6.4).

256 6.3. Parity, and the nine-component block

257 The cube is centrosymmetric. A moment that couples a field quantity of
 258 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 259 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (12) contains the first gradient alone: ∂u is
 260 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 261 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 262 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 263 by Eq. (12), and *no treatment of the second-gradient block (13) can change*
 264 *it*. What Eq. (13) contributes reaches the far field only through the $O((kh)^2)$
 265 coupling in Eq. (11). That is the order, and the origin (the variation of the
 266 field across the voxel), of the collocation error of §7, which the first-moment
 267 voxel of §8 removes.

268 With $\Delta\rho = 0$, Eq. (12) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 269 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 270 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are

271 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (19)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (20)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (21)$$

272 The density channel is the displacement block of Eq. (11) with the second
 273 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 274 of $\mathbf{G}$.

275 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 276 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 277 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 278 latational channel, Eq. (19), is the sphere's exactly: its eigenvalue involves
 279 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 280 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 281 the shape. The cube's shear channels differ from each other. Their isotropic
 282 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (22)$$

283 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 284 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 285 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 286 and the diagonal shear is the more depolarised. At the background of §7

287 $(\nu = 7/32)$, $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 288 0.4960.

289 *Where Eshelby enters.* Equations (19)–(21) follow from the moment integrals
 290 and the group theory alone. No ellipsoidal result is used, and no δ term is
 291 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 292 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 293 the uniformity theorem that ansatz is exact if and only if the inclusion is
 294 an ellipsoid. For a sphere the construction is therefore exact in the static
 295 limit, and for a cube it is an approximation. The hierarchy (11)–(13) states
 296 that approximation as the omitted blocks, rather than as an unquantified
 297 shape correction. A polynomial expansion about the centre cannot represent
 298 the edge and corner concentrations of an isolated cube's internal field at any
 299 finite degree. A voxel in a uniform layer has no such concentrations: its
 300 neighbours have the same properties, so its internal field is the continuum's,
 301 which is smooth. That is why a low-degree internal field suffices in a lattice
 302 (§§7–8) although it is only an approximation for an isolated cube.

303 *Against the exact sphere.* These statements are tested against the exact elas-
 304 tic Mie solution for a sphere, with no Eshelby formula in between. The
 305 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 306 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 307 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 308 route of Eq. (17) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 309 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 310 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 311 isotropic shear average, Eq. (22), is the depolarisation read from Mie's E_2

312 to 10^{-8} . The same moment taken as a principal value, over the ball with
 313 a small sphere excluded around the origin, has zero depolarisation in every
 314 channel. The whole static response of the sphere is the δ term.

315 At finite frequency the closure uses the dynamic ball moments, which are
 316 in closed form. It departs from Mie at second order in ka and first order in
 317 the contrast, and the departure is closed-form too: the ball's moments are
 318 elementary, and Mie's low-frequency series with the contrast kept exact gives
 319 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (23)$$

320 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 321 the radiation damping, agree exactly in all three channels. So the closure's
 322 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 323 own. That is the order the hierarchy assigns to the omitted second gradi-
 324 ents (§6.3), and it has the same origin, the variation of the field across the
 325 scatterer, as the collocation error of §7. Eq. (23) is derived symbolically
 326 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 327 the numerical Mie solve and an independent implementation of the closure
 328 to 5×10^{-4} , which is the size of the next order in the contrast.

329 6.4. The collocation closure

330 The collocation closure takes the internal field of a voxel as uniform, the
 331 truncation of §6.3 for the density and stiffness channels together. The voxel's

own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (24)$$

With the kernel (10) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (25)$$

an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
exact cell integral of the one-dimensional continuum kernel (Eqs. (8), (9)), so
 Eq. (25) is precisely the collocation of the continuum Lippmann–Schwinger
 equation with a strain piecewise constant over voxels. It does not contain
 $\mathbf{S}_{\text{self}}$ at all.

The package’s cube T -matrix is this closure statically: both blocks agree
 with Eq. (24) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 through its far-field form-factor corrections. Its self integrals are exact in the
 static and radiation parts but omit the real even-power terms of the Green’s
 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 self term, whose series agrees with the closed-form elastodynamic tensor to
 10^{-14} . By Eq. (25) that error cannot enter the closure scheme.

350 7. Convergence, and the closed-form error

351 *Second order.* For a layer $D = 2$ m with the contrast $\Delta\lambda = 2$ GPa, $\Delta\mu =$
 352 1 GPa, $\Delta\rho = 100 \text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000 \text{ m s}^{-1}$, $\rho =$
 353 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 354 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
 355 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (25) and with the
 356 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 357 $k_S h$ alone.

358 *The leading error in closed form.* In Eq. (25) the kernel is integrated exactly;
 359 the only approximation is the internal field's variation across a voxel. At first
 360 order in the contrast the discrete scattered field is therefore a midpoint rule
 361 applied to the continuum's Born integral, with the kernel's cell integral exact.
 362 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 363 it is flat. With k the background wavenumber of the wave concerned (k_P for
 364 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 365 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (26)$$

366 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 367 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 368 and those of Eq. (29) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 369 At the full contrast the reflection constant is unchanged (0.124 against $1/8$);
 370 the transmission constant grows to -0.092 .

371 *Why a uniform-field single site stops at second order.* The two constants have
 372 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 373 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 374 reduce the reflection error by a factor 4.45 and *increase* the transmission
 375 error by 1.7, at every n . The missing term in Eq. (26) is the first moment
 376 of the internal field over each voxel, which radiates through the derivative of
 377 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 378 channel, and a kernel whose cell integrals include the kernel's derivative.

379 8. Fourth order: the first-moment voxel

380 Let each voxel carry the mean *and* the first moment of its internal field.
 381 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_0 = 1$,
 382 $\phi_1 = s/h$, the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and
 383 strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with
 384 $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (27)$$

385 with the same two functions on every voxel gives

$$\left(\int_i \phi_a^2 \right) c_{a,i} = \int_i \phi_a w^0 + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (28)$$

386 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 387 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and
 388 the local term contributes $-\int_i \phi_a \phi_b / M_P$. Every entry is closed-form. Keeping
 389 only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth

order in both reflection and transmission (`ContinuumLimit_FourthOrder.w1`; Table 1). A single plane of first-moment voxels represents the 2 m layer

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (28). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full 9×9 system to all printed digits.

The leading error in closed form. Without scattering, the Galerkin solution is the L^2 projection Π of the incident field onto each voxel’s basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel’s cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} , $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (29)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{1, s/h\}$ for the first-moment voxel (`ContinuumLimit_`

FourthOrderError.wl). The results are in Table 2. The chain of Eq. (28)

Table 2: The leading error of the discrete layer in closed form, Eq. (29): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$

404

405 reproduces all six forms at $n = 1$ to 8 to 10^{-4} of the error, the remainder
 406 being the second order in the contrast. The first-moment voxel's leading
 407 error is the same in reflection and in transmission: its projection error is or-
 408 thogonal to everything linear across the voxel, so the leading term no longer
 409 depends on the direction in which the field is observed.

410 *In three dimensions.* The three-dimensional first-moment voxel carries the
 411 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 412 in all, tested with the same functions and coupled by cell-to-cell double in-
 413 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 414 cidence it reduces *exactly* to Eq. (28) (ContinuumLimit_Reduction3D.wl).
 415 The lateral first moments are never excited: the lattice and the incident
 416 field are symmetric under reflection about every voxel's centre planes, and
 417 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 418 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 419 ishes on the reciprocal lattice and the plane sum of the double integrals
 420 is the plate-to-plate integral in z alone. Two facts mark where genuinely

three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift, $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds only to $O(k_{\parallel} h)$.

Oblique incidence. The scheme is built spectrally. By Poisson summation the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (30)$$

with L_a the lateral Legendre factors. The whole-space tensor decays as $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma + i\beta \text{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 3). The mean-only voxel is second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel} h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (28) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (28) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes

Table 3: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (30). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

443 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
 444 transmission, and the full 9×9 construction with an incident SH wave re-
 445 produces that one-dimensional result to all printed digits. At oblique inci-
 446 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
 are those of Table 4. The orders, and with $k = k_S$ the closed-form constants

Table 4: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

447

448 of Eqs. (26) and (29), are those of the incident P wave.

449 The spectral construction is implemented twice. The second implementa-

tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the definition of the scheme: it takes the depth dependence of the transformed kernel from an independent residue construction, obtains the local weight as a numerical large- k_z limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two depths. The raw reflected displacements of the two agree to 9×10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and 20° , $n = 1, 2, 4$, with and without the first moments. Figure 1 collects the convergence.

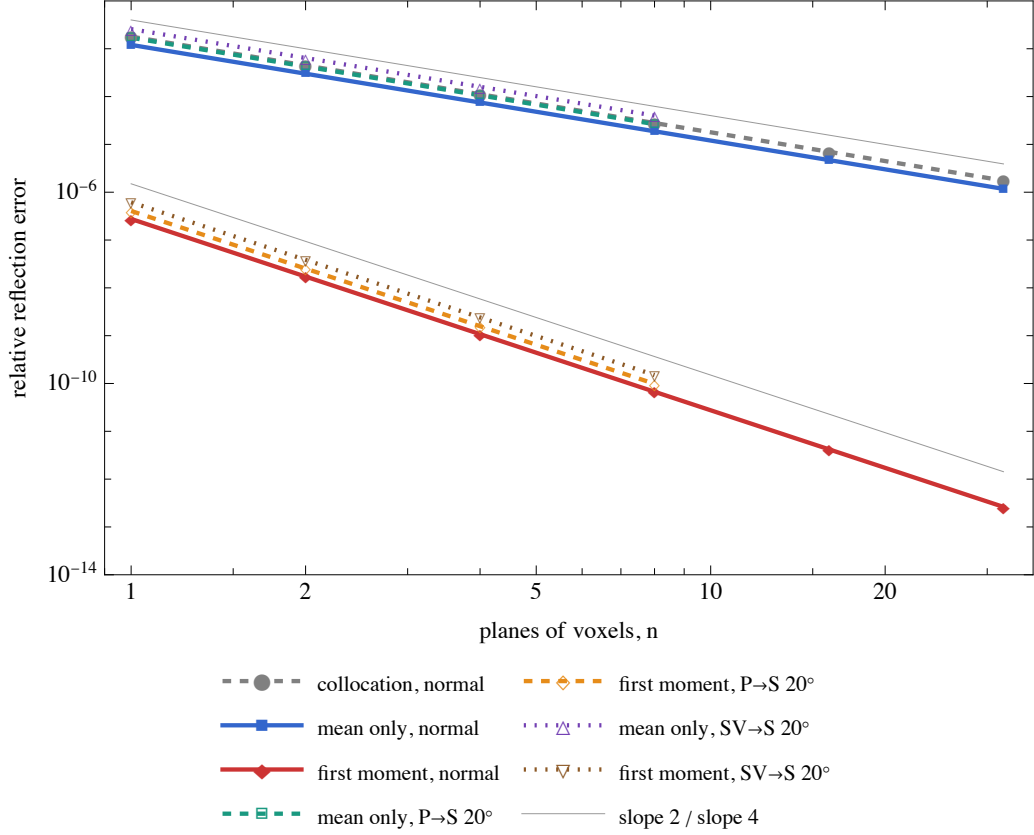


Figure 1: Relative reflection error of the discrete layer against the number of voxel planes n ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7). Filled: normal incidence, P reflection: collocation (§6), the mean-only voxel and the first-moment voxel of Eq. (28). Dashed, open squares and diamonds: P incident at 20° , the converted S wave, mean-only (green) and first-moment (orange) voxels of Eq. (30). Dotted, open triangles: SV incident at 20° , the reflected S wave, mean-only (purple) and first-moment (brown). Grey: slopes 2 and 4.

Alt text: A log-log plot of relative reflection error (vertical axis, 10^{-14} to 10^{-2}) against the number of voxel planes (horizontal axis, 1 to 32). The collocation and mean-only curves fall as straight lines of slope -2 from about 10^{-3} at one plane; the first-moment curves fall as straight lines of slope -4 from about 10^{-6} at one plane to about 10^{-13} at 32 planes. The oblique P-to-S and incident-SV curves lie alongside the normal-incidence curves of the same scheme, and grey guide lines show slopes 2 and 4.

459 9. A heterogeneous layer

460 A heterogeneous medium is not a layer of identical cells. The simplest
 461 heterogeneous case with an exact answer is a stratified model built from layers
 462 of constant properties, with real interfaces between them: the form of many
 463 layered crustal velocity models. The last test takes such a model. A layer
 464 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
 465 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
 466 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
 467 stratification with physical contrasts. The voxels refine the model's layering,
 468 with m planes of voxels in each model layer, so no voxel straddles a model
 469 interface: the natural discretisation when the model's interfaces are known.
 470 At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
 471 reverberates between the model interfaces. The exact answer is the product
 472 of the model layers' propagators. Its same-type reflection coefficients agree
 473 with those of the reflection–transmission recursion of Kennett [25], computed
 474 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
 475 now changing from plane to plane; nothing else changes. The errors are in
 476 Table 5.

477 *The interfaces cost nothing.* Every order is the one measured for the uniform
 478 layer. At a model interface the strain of the true field jumps. A voxel basis
 479 that is discontinuous from voxel to voxel represents that jump exactly when
 480 the interface lies on a voxel boundary, so the interface adds no error of its
 481 own. The only requirement is that voxel boundaries coincide with the model's
 482 interfaces. A voxel that straddled an interface would carry a contrast that is
 483 not uniform within it, a case not examined here.

Table 5: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

484 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
485 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
486 layer. The first-moment voxel needs 8, one per model layer: the model's own
487 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
488 against 36), that is 3123 against 288 per vertical column of voxels, a factor
489 of eleven. In three dimensions, with the lateral grid refined as the vertical
490 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
491 unknowns by about 2×10^4 . The same model has been solved independently
492 in Python, with a contrast for each plane: the raw reflected displacements
493 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
494 at 20° , with and without the first moments. Lateral heterogeneity needs the
495 full three-dimensional solver; it is the first extension discussed below.

496 10. Discussion

497 *Where the error of a voxel scheme lives.* The results separate the two ap-
 498 proximations a voxel scheme makes. The *coupling* between voxels makes
 499 none: averaged over the source cell it is the continuum’s cell integral ex-
 500 actly, between planes by the form factor’s zeros and within a plane by the
 501 tiling identity, and the self term it omits is exactly the one the single site
 502 carries. Every error therefore comes from the *single site*, from the internal
 503 field it assumes. For a uniform field that error is the midpoint rule’s, known
 504 in closed form; for a linear field it is the next term of the same expansion.
 505 The converse is the practical warning. A kernel that is not the cell integral
 506 (the point Green’s tensor, or a cell average that does not match the single
 507 site’s closure) leaves a local term that does not vanish as the voxels shrink
 508 (§4): the evanescent lattice orders carry a static contribution of fixed size
 509 per voxel volume. A scheme built that way converges to the wrong limit,
 510 and refinement cannot reveal it, because the error is scale-invariant.

511 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 512 single site from a modelling choice into a derived object with a stated error.
 513 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 514 only by the cubic split of Eq. (22). Its dynamic part departs from exact
 515 scattering by a sphere only through the real $(ka)^2$ term of Eq. (23); the
 516 radiation damping is exact. That term and the collocation error of Eq. (26)
 517 have the same origin, the variation of the field across the scatterer, and the
 518 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 519 field a voxel must represent is the smooth continuum field, not the field of
 520 an isolated cube, with its concentrations at the edges and corners.

521 *Relation to the discrete dipole approximation.* The same partition appears in
 522 the electromagnetic literature as the choice of polarisability. The Clausius–
 523 Mossotti relation, the lattice dispersion relation [6] and the integrated Green
 524 tensor [7] all adjust the single site so that the lattice reproduces the con-
 525 tinuum. For space-filling cubes the adjustment is neither needed nor a free
 526 parameter: the tiling identity fixes the local field exactly, the self term of the
 527 collocation closure is the one that makes it cancel, and what is left is a closed-
 528 form error rather than an empirical one. The second-order convergence re-
 529 ported for the DDA by Yurkin et al. [1] corresponds to the uniform-field
 530 voxel’s, and the opposite-signed constants of Eq. (26) explain why a polaris-
 531 ability correction can improve one observable only at the expense of another,
 532 as the package’s own form-factor corrections do in reflection and transmis-
 533 sion. Going beyond second order requires a richer internal field, which is the
 534 point of higher-order bases for volume integral equations; the first-moment
 535 voxel is the lowest such basis, and in its spectral form the Galerkin double
 536 average makes the lattice sum converge without Ewald splitting.

537 *The cost of fourth order.* A first-moment voxel carries four times as many
 538 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 539 forms show what that buys. To reach an error ε in reflection, the uniform
 540 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 541 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 542 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 543 number of unknowns by about 5×10^4 (these are the Born-order constants;
 544 at the contrast studied the first-moment voxel’s measured errors lie within
 545 5% of them, while the uniform voxel’s transmission constant grows with the

546 contrast, so the comparison is if anything conservative). The stratified model
 547 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 548 moment voxels for the same accuracy. The gain is largest precisely where it
 549 matters: when the model varies on scales that are coarse compared with the
 550 accuracy the wavefield must have.

551 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 552 mogeneous layer and a randomly stratified one, for which the discrete system
 553 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 554 moment voxel in a full three-dimensional solver, where the gradient channels
 555 couple through the evanescent lattice orders (§8) and the tiling identity holds
 556 only to $O(k_{\parallel}h)$. The numerical evidence is at moderate, physically typical
 557 contrasts (up to 15% in density and moduli) and at $k_{\parallel}h \leq 0.3$, the range in
 558 which the single-site T -matrix is validated; the closed forms are first order
 559 in the contrast, and the departures from them at the full contrast are of
 560 second order. What is established is the *order* of convergence and its lead-
 561 ing constant, not the behaviour at strong contrast, where the single site's
 562 own closure may limit the accuracy before the lattice coupling does. The
 563 single site's dynamic error is in closed form for a sphere only; for a cube,
 564 for which no exact solution exists, it is known through the layer (§§7–8).
 565 Two extensions follow directly. The first is the three-dimensional solver it-
 566 self. The second is a stratified *background*, a layered reference medium, as
 567 distinct from the stratified contrast of §9. There the lattice coupling then
 568 carries the reverberation of the layering, which is not translation invariant in
 569 depth, and whether a tiling identity holds for it is the natural next question.

570 11. Conclusions

571 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
572 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
573 tween voxels is the continuum’s cell integral exactly, and at normal incidence
574 the lattice kernel is available in closed form with no lattice sum. The single
575 site is derived rather than assumed. The moments of the Green’s tensor over
576 the cube, taken as distributions, give a closed hierarchy whose first-gradient
577 truncation is the uniform-strain closure. That closure is exact for a sphere
578 in the static limit, and it departs from exact scattering by a sphere only by a
579 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
580 the sphere only by a cubic anisotropy of the shear response. The collocation
581 closure carries the same self term as the lattice kernel, which cancels, so the
582 discrete system is a collocation of the continuum equation. It converges at
583 second order, with a leading error that is a midpoint-rule error of the internal
584 field, known in closed form for every discretisation, and opposite in sign in
585 reflection and transmission. A voxel that also carries the first moment of
586 its internal field converges at fourth order, at normal incidence (where the
587 three-dimensional scheme reduces exactly to one dimension) and at oblique
588 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
589 direction of observation. The orders survive the interfaces of a randomly
590 stratified model, provided voxel boundaries coincide with its interfaces, and
591 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel per
592 model layer, where the uniform-field voxel needs about forty. The results,
593 and the evidence for each, are collected in Table 6.

Table 6: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (9)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (10)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (17)	sum rule (18) exact; M, Q equal an independent archive, 81/81
Channels (19)–(21)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (22)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube’s bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (23)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (25)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (26)	$n = 1\text{--}16$, to 10^{-4} of the error
First-moment voxel, Eq. (28)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (30)	order 4, P and S, 20° and 40°
Incident SV and SH, all schemes	orders 2 and 4; constants with k_S to 4×10^{-4}

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596 the Research School of Earth Sciences of the Australian National University,
597 Canberra [9].

598 **Declaration of competing interest**

599 The author declares no known competing financial interests or personal
600 relationships that could have appeared to influence the work reported in this
601 paper.

602 **Data availability**

603 The Mathematica notebooks and Python scripts that derive and check ev-
604 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
605 5281/zenodo.23003241 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
606 MultipleScatteringCalculations.

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